

Лабораторная работа №2

Настройка DNS-сервера

Агаева Зейнаб Фуад кызы

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1. Цели и задачи работы
2. Процесс выполнения лабораторной работы
3. Выводы по проделанной работе

1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Получить практические навыки установки, настройки и диагностики DNS-сервера BIND на виртуальной машине **server**.

2. 2. Процесс выполнения лабораторной работы

2.1 Устанавливаю BIND и проверяю DNS-запрос

```
Installed:
  bind-32:9.18.33-15.el10_2.10.x86_64          bind-dnssec-utils-32:9.18.33-15.el10_2.10.x86_64

Complete!
[root@server.zfagaeva.net ~]# dig www.yandex.ru

; <<> DiG 9.18.33 <<> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 36842
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                466     IN      A      77.88.55.88
www.yandex.ru.                466     IN      A      5.255.255.77
www.yandex.ru.                466     IN      A      77.88.44.55

;; Query time: 6 msec
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
;; WHEN: Fri Sep 11 08:06:51 UTC 2026
;; MSG SIZE rcvd: 90

[root@server.zfagaeva.net ~]#
```

Рисунок 1: Установка BIND и проверка разрешения имени www.yandex.ru

2.2 Анализирую resolv.conf и named.conf

```
-----
[root@server.zfagaeva.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search zfagaeva.net
nameserver 192.168.1.1
[root@server.zfagaeva.net ~]# cat /etc/named.conf
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 127.0.0.1; };
    listen-on-v6 port 53 { ::1; };
    directory      "/var/named";
    dump-file       "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file   "/var/named/data/named.secroots";
    recursing-file  "/var/named/data/named.recursing";
    allow-query     { localhost; };

/*
- If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
- If you are building a RECURSIVE (caching) DNS server, you need to enable
  recursion.
- If your recursive DNS server has a public IP address, you MUST enable access
  control to limit queries to your legitimate users. Failing to do so will
  cause your server to become part of large scale DNS amplification
*/
}
```

Рисунок 2: Анализ файлов resolv.conf и named.conf

2.3 Анализирую файл named.ca

```
[root@server.zfagaeva.net ~]#  
[root@server.zfagaeva.net ~]# cat /var/named/named.ca  
;  
; This file holds the information on root name servers needed to  
; initialize cache of Internet domain name servers  
; (e.g. reference this file in the "cache . <file>"  
; configuration file of BIND domain name servers).  
;  
;  
; This file is made available by InterNIC  
; under anonymous FTP as  
; file /domain/named.cache  
; on server FTP.INTERNIC.NET  
; -OR- RS.INTERNIC.NET  
;  
;  
; last update: December 20, 2023  
; related version of root zone: 2023122001  
;  
;  
; FORMERLY NS.INTERNIC.NET  
;  
;  
; 3600000 NS A.ROOT-SERVERS.NET.  
A.ROOT-SERVERS.NET. 3600000 A 198.41.0.4  
A.ROOT-SERVERS.NET. 3600000 AAAA 2001:503:ba3e::2:30  
;  
;  
; FORMERLY NS1.ISI.EDU  
;  
;  
; 3600000 NS B.ROOT-SERVERS.NET.  
B.ROOT-SERVERS.NET. 3600000 A 170.247.170.2  
B.ROOT-SERVERS.NET. 3600000 AAAA 2801:1b8:10::b  
;  
;  
; FORMERLY C.PSI.NET  
;  
;  
; 3600000 NS C.ROOT-SERVERS.NET.
```

Рисунок 3: Просмотр файла корневых DNS-серверов named.ca

2.4 Анализирую стандартные локальные зоны

```
[root@server.zfagaeva.net ~]#  
[root@server.zfagaeva.net ~]# cat /var/named/named.localhost  
$TTL 1D  
@      IN SOA  @ rname.invalid. (   
                                0      ; serial  
                                1D     ; refresh  
                                1H     ; retry  
                                1W     ; expire  
                                3H )   ; minimum  
      NS      @  
      A      127.0.0.1  
      AAAA    ::1  
[root@server.zfagaeva.net ~]# cat /var/named/named.loopback  
$TTL 1D  
@      IN SOA  @ rname.invalid. (   
                                0      ; serial  
                                1D     ; refresh  
                                1H     ; retry  
                                1W     ; expire  
                                3H )   ; minimum  
      NS      @  
      A      127.0.0.1  
      AAAA    ::1  
      PTR     localhost.  
[root@server.zfagaeva.net ~]#
```

Рисунок 4: Просмотр файлов named.localhost и named.loopback

2.5 Запускаю службу named

```
[root@server.zfagaeva.net ~]# systemctl start named
[root@server.zfagaeva.net ~]# systemctl enable named
Created symlink '/etc/systemd/system/multi-user.target.wants/named.service' → '/usr/lib/systemd/system/named.service'.
[root@server.zfagaeva.net ~]# dig @127.0.0.1 www.yandex.ru

; <<>> DiG 9.18.33 <<>> @127.0.0.1 www.yandex.ru
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 6030
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 60568b2dfbc78b86010000006aa3b7b5dafec2fc60b7b40a (good)
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                600     IN      A      77.88.55.88
www.yandex.ru.                600     IN      A      77.88.44.55
www.yandex.ru.                600     IN      A      5.255.255.77

;; Query time: 83 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Fri Sep 11 08:11:33 UTC 2026
;; MSG SIZE rcvd: 118

[root@server.zfagaeva.net ~]#
```

Рисунок 5: Запуск named и проверка локального кэширующего DNS-сервера

2.6 Назначаю локальный DNS-сервер

```
-----
[root@server.zfagaeva.net ~]# nmcli connection edit eth0

===| nmcli interactive connection editor |===

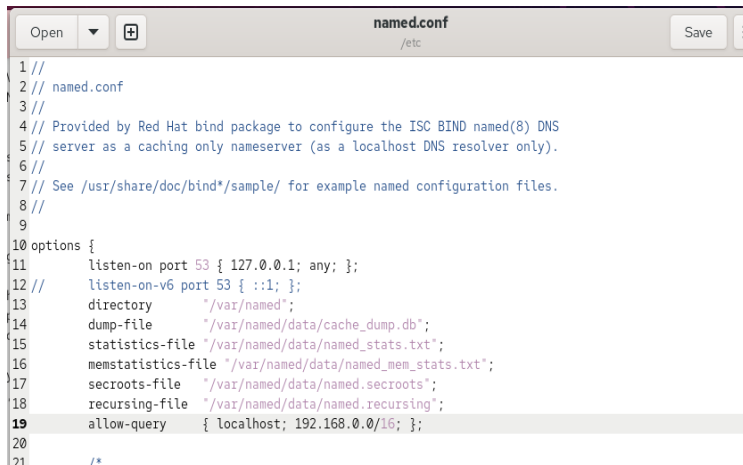
Editing existing '802-3-ethernet' connection: 'eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-1x, dcb, sriov, ethtool, match
, ipv4, ipv6, prefix-delegation, hostname, link, tc, proxy
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'eth0' (31ef6182-2c55-44b3-8944-a2af7d10d59a) successfully updated.
nmcli> quit
[root@server.zfagaeva.net ~]# systemctl restart NetworkManager
[root@server.zfagaeva.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search zfagaeva.net
nameserver 127.0.0.1
[root@server.zfagaeva.net ~]#
```

Рисунок 6: Настройка DNS-сервера по умолчанию через NetworkManager

2.7 Настраиваю named.conf для внутренней сети



```
named.conf
/etc

1 //
2 // named.conf
3 //
4 // Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
5 // server as a caching only nameserver (as a localhost DNS resolver only).
6 //
7 // See /usr/share/doc/bind*/sample/ for example named configuration files.
8 //
9
10 options {
11     listen-on port 53 { 127.0.0.1; any; };
12 //     listen-on-v6 port 53 { ::1; };
13     directory         "/var/named";
14     dump-file          "/var/named/data/cache_dump.db";
15     statistics-file     "/var/named/data/named_stats.txt";
16     memstatistics-file  "/var/named/data/named_mem_stats.txt";
17     secroots-file       "/var/named/data/named.secroots";
18     recursing-file      "/var/named/data/named.recursing";
19     allow-query         { localhost; 192.168.0.0/16; };
20
21     /*
```

Рисунок 7: Изменение параметров listen-on и allow-query

2.8 Настраиваю межсетевой экран

```
[root@server.zfagaeva.net ~]#  
[root@server.zfagaeva.net ~]# gedit /etc/named.conf  
[root@server.zfagaeva.net ~]# firewall-cmd --add-service=dns  
success  
[root@server.zfagaeva.net ~]# firewall-cmd --add-service=dns --permanent  
success  
[root@server.zfagaeva.net ~]# lsof | grep UDP  
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs  
Output information may be incomplete.  
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc  
Output information may be incomplete.  
avahi-dae 1060          avahi 12u  IPv4           10296      0t0      UDP *:mdns  
avahi-dae 1060          avahi 13u  IPv6           10297      0t0      UDP *:mdns  
chronyd  1061          chrony 4u   IPv4           10344      0t0      UDP localhost:323  
chronyd  1061          chrony 5u   IPv6           10345      0t0      UDP localhost:323  
named    15874          named 25u  IPv4           59889      0t0      UDP localhost:domain  
named    15874          named 26u  IPv4           59890      0t0      UDP localhost:domain  
named    15874          named 31u  IPv6           59893      0t0      UDP localhost:domain  
named    15874          named 32u  IPv6           59894      0t0      UDP localhost:domain  
named    15874 15875 isc-net-0    named 25u  IPv4           59889      0t0      UDP localhost:domain  
named    15874 15875 isc-net-0    named 26u  IPv4           59890      0t0      UDP localhost:domain  
named    15874 15875 isc-net-0    named 31u  IPv6           59893      0t0      UDP localhost:domain  
named    15874 15875 isc-net-0    named 32u  IPv6           59894      0t0      UDP localhost:domain  
named    15874 15876 isc-net-0    named 25u  IPv4           59889      0t0      UDP localhost:domain  
named    15874 15876 isc-net-0    named 26u  IPv4           59890      0t0      UDP localhost:domain  
named    15874 15876 isc-net-0    named 31u  IPv6           59893      0t0      UDP localhost:domain  
named    15874 15876 isc-net-0    named 32u  IPv6           59894      0t0      UDP localhost:domain  
named    15874 15877 isc-net-0    named 25u  IPv4           59889      0t0      UDP localhost:domain  
named    15874 15877 isc-net-0    named 26u  IPv4           59890      0t0      UDP localhost:domain  
named    15874 15877 isc-net-0    named 31u  IPv6           59893      0t0      UDP localhost:domain  
named    15874 15877 isc-net-0    named 32u  IPv6           59894      0t0      UDP localhost:domain  
named    15874 15878 isc-net-0    named 25u  IPv4           59889      0t0      UDP localhost:domain  
named    15874 15878 isc-net-0    named 26u  IPv4           59890      0t0      UDP localhost:domain
```

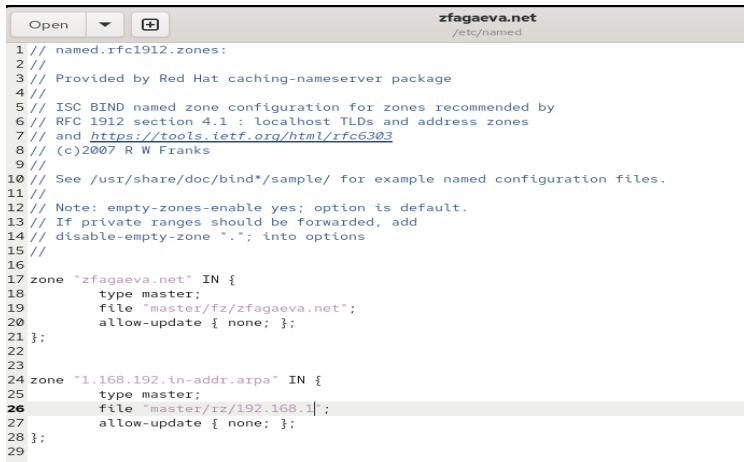
Рисунок 8: Разрешение DNS-трафика и проверка UDP-портов

2.9 Подключаю файл описания DNS-зон

```
44
45 logging {
46     channel default_debug {
47         file "data/named.run";
48         severity dynamic;
49     };
50 };
51
52 zone "." IN {
53     type hint;
54     file "named.ca";
55 };
56
57 include "/etc/named.rfc1912.zones";
58 include "/etc/named.root.key";
59 include "/etc/named/zfagaeva.net";
```

Рисунок 9: Добавление include в файл named.conf

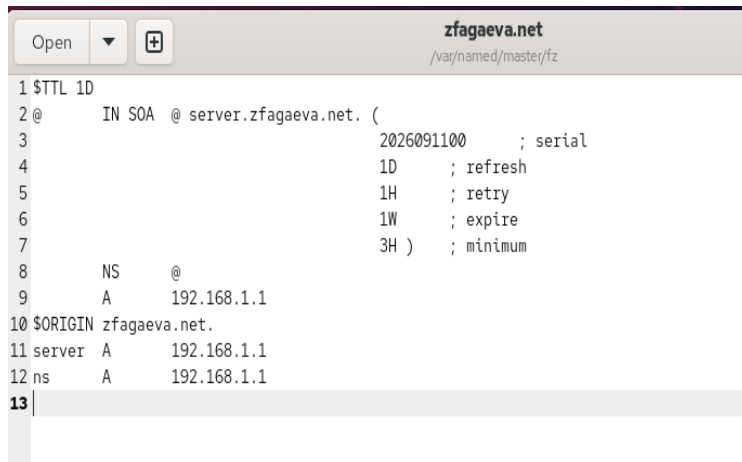
2.10 Описываю прямую и обратную зоны



```
1 // named.rfc1912.zones:
2 //
3 // Provided by Red Hat caching-nameserver package
4 //
5 // ISC BIND named zone configuration for zones recommended by
6 // RFC 1912 section 4.1 : localhost TLDs and address zones
7 // and https://tools.ietf.org/html/rfc6303
8 // (c)2007 R W Franks
9 //
10 // See /usr/share/doc/bind*/sample/ for example named configuration files.
11 //
12 // Note: empty-zones-enable yes; option is default.
13 // If private ranges should be forwarded, add
14 // disable-empty-zone "."; into options
15 //
16
17 zone "zfagaeva.net" IN {
18     type master;
19     file "master/fz/zfagaeva.net";
20     allow-update { none; };
21 };
22
23
24 zone "1.168.192.in-addr.arpa" IN {
25     type master;
26     file "master/rz/192.168.1";
27     allow-update { none; };
28 };
29
```

Рисунок 10: Описание зон zfagaeva.net и 1.168.192.in-addr.arpa

2.11 Настраиваю прямую DNS-зону

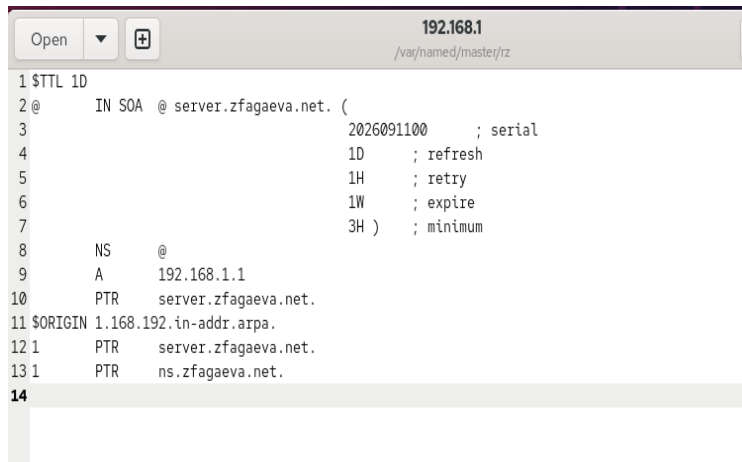


The screenshot shows a text editor window titled "zfagaeva.net" with a subtitle "/var/named/master/fz". The editor contains a DNS zone file for the domain "zfagaeva.net". The file is numbered 1 through 13 on the left margin. The content of the file is as follows:

```
1 $TTL 1D
2 @      IN SOA  @ server.zfagaeva.net. (
3                               2026091100      ; serial
4                               1D              ; refresh
5                               1H              ; retry
6                               1W              ; expire
7                               3H )            ; minimum
8      NS      @
9      A      192.168.1.1
10 $ORIGIN zfagaeva.net.
11 server A    192.168.1.1
12 ns     A    192.168.1.1
13
```

Рисунок 11: Файл прямой зоны zfagaeva.net

2.12 Настраиваю обратную DNS-зону



The screenshot shows a graphical user interface for editing a DNS zone file. At the top, there is a header bar with a title "192.168.1" and a file path "/var/named/master/rz". Below the header, the zone file content is displayed in a text area. The content is a BIND zone file for a reverse DNS zone. It starts with a TTL of 1D, followed by an SOA record for the zone. Then, there are NS, A, and PTR records. The PTR records are for the IP addresses 192.168.1.1 and 192.168.1.2. The file ends with a line number 14.

```
1 $TTL 1D
2 @      IN SOA  @ server.zfagaeva.net. (
3                                     2026091100      ; serial
4                                     1D              ; refresh
5                                     1H              ; retry
6                                     1W              ; expire
7                                     3H )            ; minimum
8      NS   @
9      A    192.168.1.1
10     PTR   server.zfagaeva.net.
11 $ORIGIN 1.168.192.in-addr.arpa.
12 1       PTR   server.zfagaeva.net.
13 1       PTR   ns.zfagaeva.net.
14
```

Рисунок 12: Файл обратной зоны 192.168.1

2.13 Настраиваю права доступа и SELinux

```
[root@server.zfagaeva.net rz]#  
[root@server.zfagaeva.net rz]# chown -R named:named /etc/named  
[root@server.zfagaeva.net rz]# chown -R named:named /var/named  
[root@server.zfagaeva.net rz]# restorecon -vR /etc  
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0  
Relabeled /etc/named.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:named_conf_t:s0  
[root@server.zfagaeva.net rz]# restorecon -vR /var/named/  
[root@server.zfagaeva.net rz]# getsebool -a | grep named  
named_tcp_bind_http_port --> off  
named_write_master_zones --> on  
[root@server.zfagaeva.net rz]# systemctl restart named  
[root@server.zfagaeva.net rz]#  
[root@server.zfagaeva.net rz]#
```

Рисунок 13: Изменение владельца файлов, восстановление контекстов SELinux и перезапуск named

2.14 Проверяю DNS-запись через dig

```
[root@server.zfagaeva.net rz]# dig ns.zfagaeva.net

; <<>> DiG 9.18.33 <<>> ns.zfagaeva.net
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 59954
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: 8009ada6c8f7bdda010000006aa3bc4a864fcdfe926cffbe (good)
;; QUESTION SECTION:
;ns.zfagaeva.net.                IN      A

;; ANSWER SECTION:
ns.zfagaeva.net.                86400   IN      A      192.168.1.1

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Fri Sep 11 08:31:06 UTC 2026
;; MSG SIZE rcvd: 88

[root@server.zfagaeva.net rz]#
```

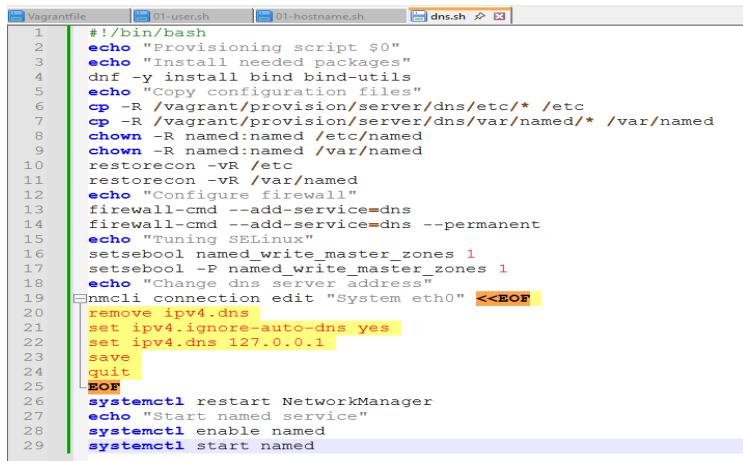
Рисунок 14: Проверка записи ns.zfagaeva.net

2.15 Проверяю работу DNS-сервера через host

```
root@server.zfagaeva.net rz#  
[root@server.zfagaeva.net rz]# host -l zfagaeva.net  
zfagaeva.net name server zfagaeva.net.  
zfagaeva.net has address 192.168.1.1  
ns.zfagaeva.net has address 192.168.1.1  
server.zfagaeva.net has address 192.168.1.1  
[root@server.zfagaeva.net rz]# host -a zfagaeva.net  
Trying "zfagaeva.net"  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 17050  
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 0  
  
;; QUESTION SECTION:  
;zfagaeva.net. IN ANY  
  
;; ANSWER SECTION:  
zfagaeva.net. 86400 IN SOA zfagaeva.net. server.zfagaeva.net. 2026091100 86400 3600 604800 10800  
zfagaeva.net. 86400 IN NS zfagaeva.net.  
zfagaeva.net. 86400 IN A 192.168.1.1  
  
Received 103 bytes from 127.0.0.1#53 in 0 ms  
[root@server.zfagaeva.net rz]# host -t A zfagaeva.net  
zfagaeva.net has address 192.168.1.1  
[root@server.zfagaeva.net rz]# host -t PTR 192.168.1.1  
1.1.168.192.in-addr.arpa domain name pointer ns.zfagaeva.net.  
1.1.168.192.in-addr.arpa domain name pointer server.zfagaeva.net.  
[root@server.zfagaeva.net rz]#
```

Рисунок 15: Проверка прямого и обратного разрешения имён

2.16 Создаю скрипт dns.sh



```
1  #!/bin/bash
2  echo "Provisioning script $0"
3  echo "Install needed packages"
4  dnf -y install bind bind-utils
5  echo "Copy configuration files"
6  cp -R /vagrant/provision/server/dns/etc/* /etc
7  cp -R /vagrant/provision/server/dns/var/named/* /var/named
8  chown -R named:named /etc/named
9  chown -R named:named /var/named
10 restorecon -vR /etc
11 restorecon -vR /var/named
12 echo "Configure firewall"
13 firewall-cmd --add-service=dns
14 firewall-cmd --add-service=dns --permanent
15 echo "Tuning SELinux"
16 setsebool named_write_master_zones 1
17 setsebool -P named_write_master_zones 1
18 echo "Change dns server address"
19 nmcli connection edit "System eth0" <<EOF
20 remove ipv4.dns
21 set ipv4.ignore-auto-dns yes
22 set ipv4.dns 127.0.0.1
23 save
24 quit
25 EOF
26 systemctl restart NetworkManager
27 echo "Start named service"
28 systemctl enable named
29 systemctl start named
```

Рисунок 16: Скрипт автоматической настройки DNS-сервера

3. 3. Выводы по проделанной работе

В ходе лабораторной работы был установлен и настроен DNS-сервер BIND на виртуальной машине **server**.

Была выполнена настройка кэширующего DNS-сервера, а локальный адрес **127.0.0.1** был назначен DNS-сервером по умолчанию.

Для домена **zfagaeva.net** были созданы прямая и обратная DNS-зоны.

Работоспособность DNS-сервера была проверена с помощью утилит **dig** и **host**.

Также был подготовлен скрипт **dns.sh**, автоматизирующий установку и настройку DNS-сервера во внутреннем окружении виртуальной машины.